SUPPORTING ADULT GUIDANCE

Rise, run and two points

Straight Lines and Connected Representations. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner finds the **gradient of a line from two points** by building a **rise-and-run triangle** and calculating **rise ÷ run** (the change in y divided by the change in x). The emphasis is on **rise ÷ run in that order**, keeping the **sign** when a line falls, and seeing that **any two points on the same line give the same gradient**.

Driving Question: *How can two points reveal the gradient without showing the equation?*

The mathematics, in plain terms

To find a line's **gradient** from two points, you make a right-angled triangle between them. The **run** is the horizontal change — how far you move across, the change in x. The **rise** is the vertical change — how far you move up or down, the change in y. The gradient is **rise ÷ run**. For points (1, 2) and (4, 8): run = 4 – 1 = 3, rise = 8 – 2 = 6, gradient = 6 ÷ 3 = 2. The order matters: it is rise *over* run, not run over rise. The key idea of the lesson is that the gradient **belongs to the line, not to the two points you happen to pick**: a wider triangle has a bigger rise *and* a bigger run, and rise ÷ run stays the same. When a line **falls** as x increases, the rise is **negative** — for (0, 6) and (3, 0), rise = 0 – 6 = –6, run = 3, gradient = –6 ÷ 3 = –2 — so keeping the minus sign is essential. In the real world this is any **rate** read from two measurements: litres per minute from two tank readings, pounds per kilometre from two points on a taxi meter, a battery falling so many percent per hour. The next lesson combines this gradient with the intercept to write the whole equation $y = mx + c$.

Rise over run, in that order

The heart of this lesson is a single, reliable routine: work out the run (across), work out the rise (up or down, with its sign), then divide rise by run. Two slips account for most errors, and both are worth naming out loud. The first is doing run ÷ rise by mistake — turning the fraction upside down. The second is dropping the minus sign on a falling line. Grounding the routine in a drawn triangle keeps it concrete: the learner can see the across-and-up movement rather than juggling a formula. It is worth protecting the central insight that a wider triangle gives the same gradient — pick a second pair of points on the same line and show rise ÷ run is unchanged. This is what makes 'the gradient of the line' a meaningful phrase, and it sets up writing the equation in the next lesson.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Same line, different triangle: two triangles on one line, both giving the same rise ÷ run.	~10 min

Movement	Gradient Triangle Builder: drag two points, read rise, run and gradient = rise ÷ run.	~15 min
Reflection	Right or slip?: spot the two classic errors — run/rise swapped, and a lost minus sign.	~10 min
Creativity	Rate from two readings: find a real rate (including a negative one) from two measurements.	~5 min
Rest	A pause after seeing that any two points on a line give the same gradient.	~3 min
Nutrition	Contextual (Mode A): flow rates, fares, battery drain — every 'per' is a rise ÷ run.	~2 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Dividing run by rise (turning the fraction upside down)

Repair: “Gradient is rise ÷ run. Which is the rise here, and which is the run? Write it as rise on top.”

Dropping the minus sign on a line that falls

Repair: “As x increases, does y go up or down? Down means the rise is negative — keep the minus.”

Thinking a bigger triangle gives a bigger gradient

Repair: “A wider triangle has a bigger rise and a bigger run. Work out rise ÷ run for both — what do you notice?”

Subtracting the coordinates in a different order for rise and run

Repair: “Use the same order for both: second point minus first, for the y’s and for the x’s.”

Reading rise or run straight off a point instead of the difference

Repair: “Rise and run are changes — end minus start, not the coordinate itself.”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: reading coordinates, subtracting to find a difference (including negatives), and treating a fraction as a division. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the routine and the two slips with the learner:

- “What is the run — how far across from one point to the other?”
- “What is the rise — how far up or down? Is it positive or negative?”
- “Now rise ÷ run — which number goes on top?”
- “Take a different pair of points on the same line. Do you get the same gradient?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“Does it matter which point I start from?”

No — as long as you are consistent. Do second minus first for both the rise and the run (or first minus second for both). Starting from the other point flips the sign of the rise and the run, and the two minuses cancel, so the gradient is the same.

“Why build a triangle instead of just using a formula?”

The triangle makes the rise and run visible, so the meaning stays attached to the numbers. Learners who can see the across-and-up movement make far fewer upside-down and sign errors than those reciting a formula.

“How does this set up the next lesson?”

Once a learner can find a gradient from two points, they have the m in $y = mx + c$. The next lesson pairs that gradient with the intercept to write the whole equation of the line.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.

Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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